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Q. What's the difference between first-generation Intel i5 processor and second-generation Intel i5 processor. If the generation of the processor changes, does it influence the performance of the computer?

Ram Sandeep
Andhra Pradesh

A. Core i5 is a CPU series manufactured by Intel. It is aimed at mid-range computers and mobile applications. The difference will be better understood if we go through the entire range of latest processors introduced by Intel.

First-generation processors were introduced in 2010. This generation included Core i3, eight Core i5 and five Core i7 models. Core i3 is a low-end processor. The desktop and mobile versions of these processors have dual-core technology and Intel Hyper Threading support. It does not have the Intel Turbo Boost support.

Core i5 has both dual and quad processors and at least two of its processors support the Intel Turbo Boost, graphic designs and Hyper Threading. Core i7 are the most powerful processors. These have four processors that support Hyper Threading and Turbo Boost technology. The Core i7 processor has both quad and dual cores. It is the highest-priced processor in the Core family.

Intel's second-generation processors were introduced in 2011. These are capable of superior performance, better graphics capabilities and improved energy efficiency—in laptops it means longer battery life.

These processors are based on Intel's Sandy Bridge architecture. The series contains desktop processors and 29 mobile processors. This family of processors was based on the 32nm micro architecture by Intel and was the first to integrate the processor and memory controller graphics on the same die.

The second-generation processors include video-performing encoding hardware, a video HD, 3D stereoscopic

content for television viewing, and support of full streaming HD. This family of Cores contains one Core i7 extreme edition, twelve Core i7, twelve Core i5, four Core i3 processors, and several new graphic-performance features.

Second-generation Intel Core i5 processors have the following inbuilt features:

1. Intel Turbo Boost technology 2.0, which increases the processor's frequency.
2. Intel Smart Cache, which is allocated to each processor core based



Intel Core i5 Processor

on workload. It reduces latency and improves performance.

3. Advanced Encryption Standard (AES) instructions that add hardware acceleration to AES algorithms and speed up the execution of AES applications.

4. Better 3D graphics and high-definition video performance, which is the most improved aspect of the new chips. This feature lacked in first-generation processors.

5. Quad-core processor with four-way multitasking capability, which delivers extra speed as well as security features for protection of information and data.

Answers compiled by Col (Retd) N.C. Pande.

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